Continuation Cost Breaks the Geometric Tail Mechanism

A Bellman-corrected envelope for exact zero-repair trees

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Abstract

For the quartic zero-repair pair

$$P = \left(\tfrac{1}{8}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{8}\right), \qquad Q = \left(\tfrac{7}{64}, \tfrac{5}{16}, \tfrac{5}{32}, \tfrac{5}{16}, \tfrac{7}{64}\right),$$

we study two finite-horizon quantities: the minimum expected initial reserve $\mathcal{B}_n$, and the optimal weak-1 tail constant

$$A_n := \inf_B \sup_{j\ge 1} j\, \mathbb{P}_B(J_0 \ge j).$$

The first five computed values of A_n follow an apparently exact geometric law. We show analytically why this law arises from the two lowest reserve states, and why it cannot persist.

The obstruction is a continuation-value constraint: once the child subtree must retain enough mean reserve to solve the remaining horizon, some mass is forced above reserve state 1. This yields a universal Bellman-corrected recursion coupling the maximal zero atom with $\mathcal{B}_{n-1}$. Using only an exact certified lower bound on $\mathcal{B}_5$, we obtain a strict improvement of the geometric tail envelope at every subsequent scale and an improved necessary constant for any exact infinite-tree construction with $1/j$ root tail.

1 Setup

An exact depth-n zero-repair tree is a family of probability laws $\{B_h\}_{|h|\le n}$ on $\mathbb{Z}_{\ge 0}$ satisfying

$$Q * B_h = \sum_{x=0}^{4} P(x)\, \delta_x * B_{hx}, \qquad |h| < n. \tag{1}$$

Let

$$\mathcal{B}_n := \inf \mathbb{E} B_\varnothing, \qquad \mathcal{B}_0 := 0.$$

We introduce two further extremal quantities:

$$Z_n := \sup\left\{B_\varnothing(0) : B_\varnothing \text{ is the root of an exact depth-}n\text{ tree}\right\}, \tag{2}$$

and

$$A_n := \inf\left\{\sup_{j\ge 1} j\, \mathbb{P}(J_0 \ge j) : J_0 \sim B_\varnothing,\ B_\varnothing \text{ is feasible at depth } n\right\}. \tag{3}$$

Pruning shows that both $\mathcal{B}_n$ and A_n are nondecreasing in n.

For later use, define

$$U_n := \frac{8}{11} + \frac{3}{11}\left(\frac{16}{27}\right)^n. \tag{4}$$

2 The geometric low-state mechanism

Fix a node with $r \geq 1$ steps remaining and write

$$a = B_h(0), \qquad b = B_h(1),$$

$$u_0 = B_{h0}(0), \qquad u_1 = B_{h0}(1), \qquad v_0 = B_{h1}(0).$$

Lemma 1 (Two exact low-state identities)**.** *Every exact tree satisfies*

$$7a = 8u_0, \tag{5}$$

$$27a + 7b = 8(u_0 + u_1) + 16v_0. \tag{6}$$

Proof. Take respectively the coefficient of reserve 0 in (1), and then the sum of the coefficients at reserves 0 and 1. Multiplying by 64 gives

$$7a = 8u_0$$

and

$$7a + (20a + 7b) = 8u_0 + (8u_1 + 16v_0).$$

□

Since $b \geq 0$, $u_0 + u_1 \leq 1$, and $v_0 \leq Z_{r-1}$, (6) gives

$$27a \leq 8 + 16Z_{r-1}.$$

Hence

$$\boxed{Z_r \leq \frac{8 + 16Z_{r-1}}{27}.} \tag{7}$$

Starting from $Z_0 = 1$, iteration yields

$$\boxed{Z_r \leq U_r.} \tag{8}$$

The same identities produce a tail bound.

Lemma 2 (Geometric tail envelope)**.** *For every $n \geq 1$,*

$$\boxed{A_n \geq \frac{2}{47}(19 - 16Z_{n-1}) \geq g_n,} \tag{9}$$

where

$$g_n := \frac{162}{517}\left[1 - \left(\frac{16}{27}\right)^n\right]. \tag{10}$$

Proof. Let

$$T_j := \mathbb{P}(J_0 \geq j), \qquad jT_j \leq A.$$

Then

$$a = 1 - T_1, \qquad b = T_1 - T_2,$$

and therefore

$$27a + 7b = 27 - 20T_1 - 7T_2 \geq 27 - \frac{47}{2}A.$$

On the other hand, (6) gives

$$27a + 7b \leq 8 + 16Z_{n-1}.$$

Rearranging proves the first inequality in (9); substituting (4) gives (10). □

The first five numerical tail optima coincide with g_n to solver precision. The next section explains why this low-state mechanism cannot continue indefinitely.

3 Continuation cost forces mass above state one

The root of any finite depth-r tree may be truncated at $4r$ without changing its zero atom. Indeed, in a causal realization one subtracts

$$D = (J_0 - 4r)_+$$

from the entire reserve trajectory. Since the reserve can decrease by at most 4 per step, nonnegativity is preserved.

Consequently, after the first output $X_1 = 0$, the child reserve may be assumed supported on

$$\{0, \dots, 4r + 4\}.$$

Put

$$q_0 := \mathbb{P}_{B_{h0}}(J \geq 2) = 1 - u_0 - u_1.$$

Lemma 3 (Continuation-forced high-state mass). *For every node with $r \geq 1$ steps remaining,*

$$\boxed{q_0 \geq \frac{\left(\mathcal{B}_{r-1} - 1 + \frac{7}{8}a\right)_+}{4r + 3}.} \tag{11}$$

Proof. The child following output 0 has $r - 1$ steps remaining, hence

$$\mathbb{E}B_{h0} \geq \mathcal{B}_{r-1}.$$

Its support is bounded by $4r + 4$, so

$$\mathbb{E}B_{h0} \leq u_1 + (4r + 4)q_0 = 1 - u_0 + (4r + 3)q_0.$$

Using $u_0 = \frac{7}{8}a$ from (5) proves (11). □

This is the basic collision between local tail optimization and future cost. If

$$\mathcal{B}_{r-1} > 1 - \frac{7}{8}a,$$

the child after output 0 cannot keep all of its mass at reserves 0 and 1.

4 The Bellman-corrected zero-atom recursion

Combining (6) with $b \geq 0$ and $v_0 \leq Z_{r-1}$ gives

$$27a \leq 8(1 - q_0) + 16Z_{r-1}.$$

Using (11) yields the master inequality

$$\boxed{27a \leq 8 + 16Z_{r-1} - \frac{8}{4r + 3}\left(\mathcal{B}_{r-1} - 1 + \frac{7}{8}a\right)_+.} \tag{12}$$

Theorem 1 (Bellman-corrected envelope). *For every $r \geq 1$,*

$$\boxed{Z_r \leq \Phi_r(Z_{r-1}, \mathcal{B}_{r-1}),} \tag{13}$$

where

$$\Phi_r(z, b) := \min\left\{\frac{8 + 16z}{27}, \frac{16(4r + 3)z + 32(r + 1) - 8b}{108r + 88}\right\}. \tag{14}$$

Proof. If the positive part in (12) vanishes, the first term in (14) applies.

If it is active, multiplication of (12) by $4r+3$ gives

$$(108r+88)a \le 16(4r+3)Z_{r-1} + 32(r+1) - 8\mathcal{B}_{r-1},$$

which is the second term.

At the switching point the two bounds match in the required piecewise order, so their minimum is valid for all feasible a. Taking the supremum over roots proves (13). □

Since Φ_r is increasing in its first argument and decreasing in its second, (8) gives the explicit consequence

$$\boxed{Z_r \le U_r - \frac{2\big(27\mathcal{B}_{r-1} + 14U_{r-1} - 20\big)_+}{27(27r+22)}.} \tag{15}$$

Thus the old geometric envelope is strictly improved whenever

$$\boxed{\mathcal{B}_{r-1} > \frac{20 - 14U_{r-1}}{27}.} \tag{16}$$

The threshold on the right tends to

$$\frac{4}{11}.$$

5 A permanent correction from the depth-five certificate

An independently verified exact dual certificate gives

$$\boxed{\mathcal{B}_5 \ge L_5 := \frac{207981249337}{549755813888} = 0.3783156886802317\ldots} \tag{17}$$

and

$$L_5 - \frac{4}{11} = \frac{88770487155}{6047313952768} > 0.$$

Since $\mathcal{B}_n$ is nondecreasing,

$$\mathcal{B}_{r-1} \ge L_5 > \frac{4}{11} \qquad (r \ge 6).$$

Because $U_{r-1} > 8/11$, condition (16) is therefore satisfied for every $r \ge 6$.

Corollary 1 (Permanent break of the geometric envelope)**.** *The uncorrected geometric zero-atom mechanism is strictly impossible at every scale $r \ge 6$:*

$$\boxed{Z_r < U_r, \qquad r \ge 6.} \tag{18}$$

More explicitly,

$$Z_r \le U_r - \frac{2\big(27L_5 + 14U_{r-1} - 20\big)}{27(27r+22)}. \tag{19}$$

Substitution into (9) gives, for every $n \ge 7$,

$$\boxed{A_n \ge g_n + \frac{64\big(27L_5 + 14U_{n-2} - 20\big)}{1269(27n-5)}.} \tag{20}$$

At $n = 7$ this reproduces the certified separation

$$A_7 \ge 0.3054901947038539\ldots > g_7 = 0.3053050812507245\ldots.$$

Thus the observed five-step geometric pattern cannot extend unchanged to horizon seven.

6 Recursive propagation of the correction

The correction can be propagated instead of restarting from U_r at each scale.

Set

$$\widehat{Z}_5 := U_5$$

and recursively, for $r \geq 6$,

$$\boxed{\widehat{Z}_r := \Phi_r(\widehat{Z}_{r-1}, L_5).} \tag{21}$$

Proposition 1 (Recursive Bellman envelope)**.** *For every $r \geq 5$,*

$$\boxed{Z_r \leq \widehat{Z}_r.}$$

Consequently,

$$\boxed{A_n \geq \widehat{A}_n := \frac{2}{47}\big(19 - 16\widehat{Z}_{n-1}\big), \qquad n \geq 6.} \tag{22}$$

Proof. The claim holds at $r = 5$ by $Z_5 \leq U_5$. If $Z_{r-1} \leq \widehat{Z}_{r-1}$, then $\mathcal{B}_{r-1} \geq L_5$, and monotonicity of Φ_r gives

$$Z_r \leq \Phi_r(Z_{r-1}, \mathcal{B}_{r-1}) \leq \Phi_r(\widehat{Z}_{r-1}, L_5) = \widehat{Z}_r.$$

The tail bound follows from (9). □

All quantities in (21) are rational. Direct exact iteration gives

$$\boxed{\widehat{A}_{19} = 0.3134375576615171\ldots > \frac{162}{517} = 0.3133462282398453\ldots.} \tag{23}$$

The comparison in (23) is exact; in rational arithmetic,

$$\widehat{A}_{19} - \frac{162}{517} = \frac{3515159781244978924719086604025464 5}{3848879930365198182389073930279526072 32} > 0.$$

Since A_n is nondecreasing, this yields the following consequence.

Corollary 2 (Necessary constant for a critical infinite tree)**.** *Suppose an exact infinite zero-repair tree has root reserve J_0 satisfying*

$$\mathbb{P}(J_0 \geq j) \leq \frac{A}{j} \qquad (j \geq 1).$$

Then

$$\boxed{A \geq A_{19} \geq 0.3134375576615171\ldots.} \tag{24}$$

Proof. Restricting the infinite tree to any finite horizon preserves its root law, hence

$$A \geq A_n$$

for every n. Apply (23). □

7 Interpretation

The geometric factor $16/27$ is not a numerical accident. It is the exact low-state recursion obtained when the child after output 0 can concentrate its reserve on states 0 and 1.

The continuation constraint changes this picture. A child with $r-1$ steps remaining must satisfy

$$\mathbb{E}B_{h0} \geq \mathcal{B}_{r-1}.$$

Once this exceeds the mean compatible with the low-state extremizer, positive mass above reserve 1 becomes compulsory. This mass feeds back into the two lowest convolution equations and lowers the maximal possible zero atom.

Thus the structural mechanism is

$$\boxed{\begin{array}{c}\text{local tail optimization} \longrightarrow \text{continuation-value collision} \longrightarrow \text{forced high-state mass} \\ \longrightarrow \text{Bellman-corrected tail envelope}\end{array}}$$

The result does not resolve whether

$$\mathcal{B}_n = O(\log n).$$

It does show that any critical $1/j$ infinite-tree construction must survive a continuation constraint that is invisible in the original low-state geometric recursion.

Remark 1. *The argument uses the certified depth-five mean bound only through the single inequality* $L_5 > 4/11$. *Any stronger future lower bound on* $\mathcal{B}_r$ *can be inserted directly into* (13), *producing a sharper tail envelope without changing the proof.*